SonixED™ Skid System

Model: **SONIXED-SKID**
Version: **2025–A7**

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Overview

The SonixED™ Recovery Skid integrates HydroVolta's proprietary hybrid electrodialysis technology with ultrasound antifouling and intelligent PLC automation. Engineered for high recovery and minimal maintenance, it enables chemical-free operation for industrial and groundwater applications.

General Description

The SonixED™ skid system offers a complete plug-and-play electrodialysis solution for desalination plus monovalent and divalent ion recovery. Each skid includes pre-assembled modules with pumps, manifolds, sensors, and PLC controls, providing stable performance and reliable results under varying water conditions.

System Overview

Parameter	Specification
Technology	Hybrid electrodialysis with built-in ultrasound transducers
Configuration	Skids with multi SonixED™ stacks and multi pumps
Operation	Automated PLC with manual override and RS485 communication
Applications	Mining, industrial reuse, groundwater desalination, brine valorization

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Performance Specifications

Parameter	Specification
Flowrate (per stack)	0.5 – 10 m^3/h
System Flow (per skid)	Up to 30 m^3/h
Recovery Rate	Up to 98%
Operating Pressure	0.5 – 2.5 bar
Temperature Range	5 – 55°C
Specific Energy	0.6 – 2.0 kWh/m^3

Key Components (Per Skid)

Component	Description
SonixED™ Stacks	1-3 high-efficiency SonixED stacks with titanium electrodes, HydroVolta membranes, and integrated ultrasound antifouling
Magnetic Pumps	3 inverter-driven polypropylene, stainless steel pumps, seawater compatible
Control System	PLC , HMI interface, manual/emergency controls, IP55-rated cabinet
Valves & Manifolds	3-way automated PVC valves, DN25-63 manifolds, stainless supports
Instrumentation	Inline pH, EC, flow, and pressure sensors with 4–20 mA outputs

Optional

Device	Description
Pre & Post-Treatment	Like sediment filtration, ultrafiltration, chlorination, pH correction, and nano filtration "*in case of the brine valorization*"

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Mechanical Characteristics

Parameter	Specification
Dimensions (per skid)	Approx. 3040 × 1340 mm x 2020
Frame Material	Aluminum profiles
Housing	PVC with thermoplastic spacers
Mobility	Mounted on lockable wheels

Optional Accessory

Device	Description
Cloud Multi-Parameter Water Analyzer (SUP-MPP1000)	Real-time monitoring and control with cloud telemetering of turbidity, pH, conductivity, DO, and residual chlorine. 7-inch touch display, RS485/4–20 mA output, PLC integration.

Key Advantages

- High recovery efficiency up **to 98%**
- Chemical-free antifouling using **ultrasonic technology**
- Low **energy** consumption and O&M cost
- Compact **modular** design – easy to transport and install
- **Smart** PLC automation with full data connectivity

Applications

- Groundwater desalination and mineral recovery
- Industrial process water reuse
- Mining and brine valorization
- Selective ion removal for chemical industries
- Zero Liquid Discharge (ZLD) systems

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Technical Illustrations

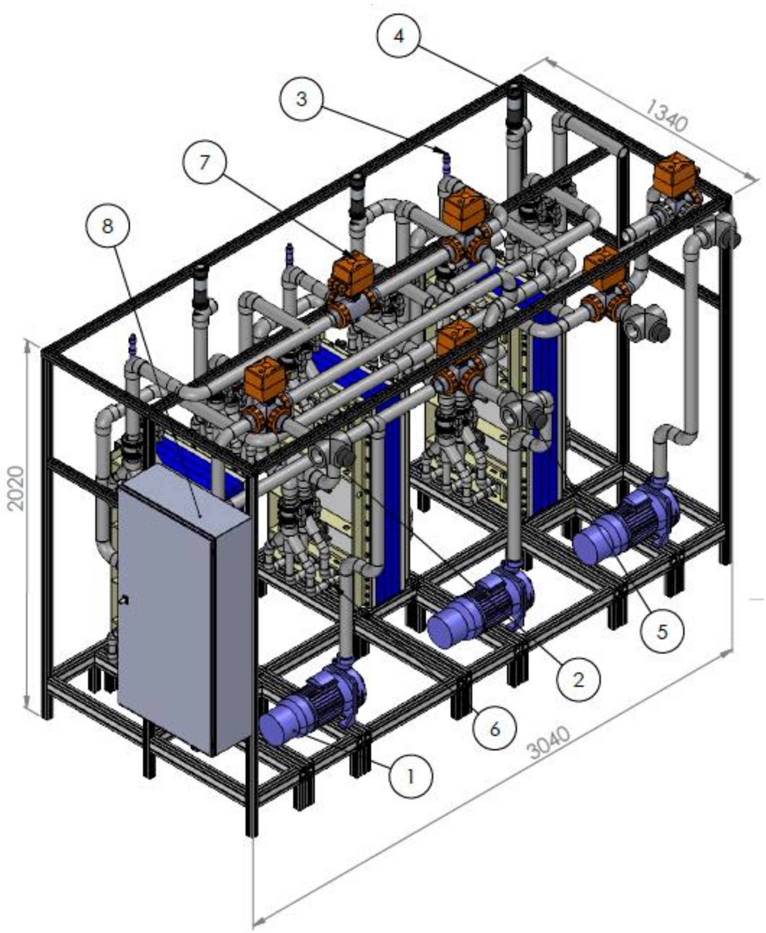

Figure 1. SonixED™ Skid – Numbered 3D View

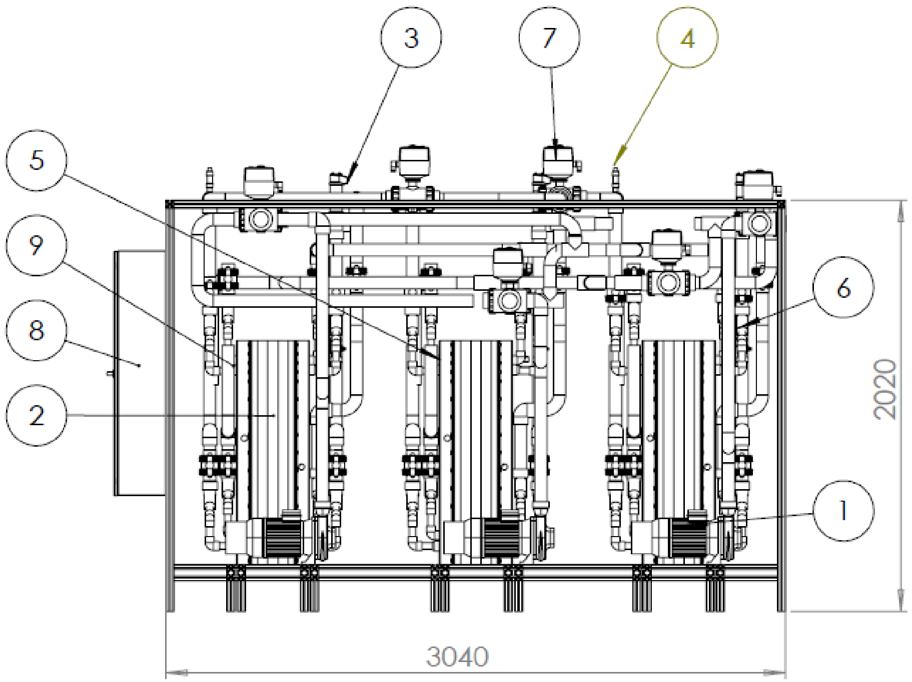

Figure 2. Dimensional Overview – Key Measurements of Skid Assembly

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Component Numbering Table

No.	Component Name	Description
1	Feed Pump	Magnetic-driven centrifugal pump for continuous feed water circulation.
2	SonixED™ Stack Module	Electrodialysis cell unit for ion separation, includes integrated ultrasound transducers.
3-4	Instrumentation Sensors	Includes inline EC, pH, and flow sensors providing real-time process data.
5	Pressure Gauge	Monitors hydraulic pressure across stacks for safety and process control.
6	Manifold Assembly	Distributes feed evenly to all stacks with PVC and stainless fittings.
7	3-Way Valve	Directs flow between feed, product, and recirculation lines for flexible operation.
8	Control Cabinet (PLC)	Houses PLC, power distribution, and HMI interface for automation and monitoring.
9	Frame & Base	Aluminum frame, providing structural support and rigidity.

Ordering Code – SonixED Skid configuration

The SonixED™ modular configuration allows customization according to process capacity, measurement density, and control sophistication.

SONIXED-SKID-A-S-M-PR-V-CB-CL		
Code	**Option**	**Description**
A	A0 / A1 / A2	Application Type: A0 = Groundwater Desalination, A1 = Industrial Water Reuse, A2 = Mining / Brine Concentration.
S	1 / 2 / 3	Number of SonixED™ stacks integrated in the skid.
M	6 / 9 / 12	pH & EC measuring points (in/out of diluate, concentrate, and electrolyte) using HydroVolta's multi-point technology.
PR	6 / 9 / 12	Pressure sensors positioned on each stream inlet and outlet.
V	2 / 4 / 6	3-way valves for reversal automation; can be configured per stack or for grouped operation.
CB	CB0 / CB1	Control box configuration: CB0 = PLC only, CB1 = PLC + touchscreen HMI.
CL	CL0 / CL1 / CL2	Cloud connectivity options: CL0 = local only, CL1 = cloud monitoring, CL2 = full cloud monitoring and control.

Example: SONIXED-SKID-A1-S3-M12-PR9-V6-CB1-CL2

Notes:

- Pumps: Always 3 per skid (feed, diluate, concentrate) — size scaled to system capacity.
- Flow Meters: Always 3 (feed, diluate outlet, concentrate outlet).
- Custom configurations available upon request.

Parameters can be customized: number of stacks, measuring points, reversal valves, and control interfaces. HydroVolta engineers can tailor each skid for pilot or industrial deployment.